ROBOTREE ROBOTICS LLP

Official Website & Platform Rebuild Cost Valuation

Target Market: India (INR / Rs.) | Assessment Date: September 2026 | Currency: Indian Rupee (INR)

1. EXECUTIVE PLATFORM VALUATION & SCOPE

This report provides a realistic, market-grounded cost analysis to architect, engineer, test, and deploy the complete Robotree Robotics LLP digital infrastructure from zero in India. The platform encompasses an 11-page public portal, multi-role user authentication, real-time administrative management suite, cryptographic certificate verification with vector PDF issuance, Razorpay payments, and a hardened TypeScript/Node.js backend with dual JSON/PostgreSQL database engines.

SENIOR FREELANCER TIER
Rs. 85,000 - Rs. 1,50,000
Timeline: 3 - 5 Weeks (Recommended)

BOUTIQUE AGENCY TIER
Rs. 1,50,000 - Rs. 2,80,000
Timeline: 6 - 10 Weeks (Team Delivery)

ENTERPRISE AGENCY TIER
Rs. 3,00,000 - Rs. 6,00,000+
Timeline: 8 - 12 Weeks (SLA & PM)

2. FREELANCER EXPERIENCE TIERS & COST BREAKDOWN IN INDIA

FREELANCER TIER	EXP.	ESTIMATED COST	DURATION	DELIVERABLE QUALITY & ARCHITECTURAL R
Junior Freelancer	1-2 Yrs	**Rs. 20,000 - Rs. 40,000**	4-8 Wks	- Relies heavily on generic WordPress themes or fragmented static HTM - High Risk: Cannot engineer custom PDF engines, TypeScript, or CSR - Typically results in insecure code, fragile data persistence, and projec
Mid-Level Full-Stack	3-5 Yrs	**Rs. 45,000 - Rs. 80,000**	4-6 Wks	- Builds standard Express/Node APIs and functional responsive UI form - Capable of handling standard CRUD, JSON databases, and basic Raz - Moderate Risk: Frequently neglects edge security (CSP, timingSafe, p
Senior Full-Stack (Recommended)	5+ Yrs	**Rs. 85,000 - Rs. 1,50,000**	3-5 Wks	- Production-grade modular TypeScript backend with robust Zod validat - Custom programmatic PDF certificate layout, timing-safe authenticatio - Low Risk: Complete automated testing, clean separation of concerns,

ALTERNATIVE FREELANCE ENGAGEMENT CONTRACTS

Hourly Engagement Model: Standard Indian Senior Full-Stack rate: Rs. 800 - Rs. 1,500/hr ($10 - $18/hr). Estimated 100 - 120 billable hours = Rs. 80

Monthly Retainer Model: Dedicated Senior Developer for 1.5 months: Rs. 60,000 - Rs. 90,000/month = Rs. 90,000 - Rs. 1,35,000 total.

Fixed Milestone Model (Safest): Milestone-based delivery with escrow release tied strictly to verified QA deliverables (detailed on Page 2).

3. FEATURE-BY-FEATURE ENGINEERING COST BREAKDOWN

MODULE / SUBSYSTEM	TECHNICAL SCOPE & DELIVERABLES	ESTIMATED MARKET COST
1. UI/UX & Responsive Pages	11 semantic HTML5 pages (Home, About, Team, Courses, Services, Gallery, Certificate, Contact, Admin, Terms, Privacy), glassmorphic design system, CSS variables, dark/light theme engine, mobile drawer.	**Rs. 35,000 - Rs. 60,000**
2. Multi-Role Portal & Auth	Role-Based Access Control (Student, Employee, Admin, Super Admin), Argon2id & PBKDF2 (210k iter) password hashing, HttpOnly session tokens, brute-force locking, and GDPR export/erasure endpoints.	**Rs. 20,000 - Rs. 35,000**
3. Admin Management Suite	Dedicated passcode login panel, full CRUD for courses (tracks, INR pricing), 17+ team members, event galleries (multi-photo uploads), inquiry leads tracking, and real-time security audit log stream.	**Rs. 30,000 - Rs. 55,000**
4. Certificate Verification Engine	Public verification pipeline by ID/QR reference code, automated vector PDF generation engine using pdf-lib with official emblem stamps, certificate download streaming, and revocation management.	**Rs. 25,000 - Rs. 45,000**
5. Course Catalog & Razorpay	Course tracks catalog, module curriculums, Razorpay payment order generation, checkout modal integration, webhook handling, and enrollment reconciliation.	**Rs. 15,000 - Rs. 25,000**
6. Enterprise Security Hardening	Helmet CSP, clickjacking frameguard (DENY), nosniff headers, CSRF double-submit token guard, same-origin validation, sliding-window rate limiters, Zod input validation, path traversal defense.	**Rs. 20,000 - Rs. 35,000**
7. DevOps & Dual DB Architecture	Dual persistence layer (Atomic JSON DB engine + PostgreSQL 16 schema migrations), Docker multi-stage build, Nginx reverse proxy configuration, systemd service, backup scripts, and 21 automated tests.	**Rs. 20,000 - Rs. 40,000**
TOTAL COMPONENT-BY-COMPONENT ENGINEERING VALUATION		**Rs. 1,65,000 - Rs. 2,95,000**

4. RECOMMENDED FREELANCER MILESTONE-BASED PAYMENT STRUCTURE

MILESTONE	KEY DELIVERABLES & VERIFICATION CHECKPOINTS	SHARE	EST. AMOUNT
Milestone 1 (Week 1-2)	Design System & Frontend Architecture: 11 responsive HTML pages, dark/light theme, interactive modals, responsive navigation, cross-browser CSS.	**30%**	**Rs. 25,000 - Rs. 35,000**
Milestone 2 (Week 2-3)	Backend APIs, Auth & Admin Suite: TypeScript REST endpoints, role authentication (Argon2/PBKDF2), and complete admin management CRUD for courses/team/events.	**35%**	**Rs. 30,000 - Rs. 45,000**
Milestone 3 (Week 3-4)	Certificate & Payment Engines: Public verification code lookup, automated server-side PDF generator (pdf-lib), Razorpay checkout & payment verification flow.	**20%**	**Rs. 20,000 - Rs. 30,000**
Milestone 4 (Week 4-5)	Security, Testing & Live Deployment: CSRF defense, rate-limiting, security headers, automated test passing, Docker/Nginx configuration, SSL cert & VPS go-live.	**15%**	**Rs. 15,000 - Rs. 20,000**

5. RECURRING OPERATIONAL & INFRASTRUCTURE EXPENSES

INFRASTRUCTURE RESOURCE	RECOMMENDED PROVIDER	FREQUENCY	ESTIMATED COST (INR)
Domain Name (.org / .in)	GoDaddy / Namecheap	Annual	**Rs. 800 - Rs. 1,500 / year**
Cloud VPS (2 vCPU, 4GB RAM)	Hetzner / DigitalOcean / AWS Lightsail	Monthly	**Rs. 600 - Rs. 1,800 / month**
Managed Database (PostgreSQL)	Supabase / Neon / Self-Hosted Docker	Monthly	**Free Tier to Rs. 1,500 / mo**
SSL Certificate (HTTPS)	Let's Encrypt / Cloudflare	Continuous	**Free (Rs. 0)**
Payment Gateway Processing	Razorpay / Cashfree Payments	Per Sale	**2.0% + GST per enrollment**
Transactional Email / SMS	Brevo / SendGrid / MSG91	Monthly	**Free Tier to Rs. 1,000 / mo**
Annual Maintenance (AMC)	Freelance DevOps / Tech Support	Optional	**Rs. 3,000 - Rs. 8,000 / month**

6. STRATEGIC RECOMMENDATIONS & COST OPTIMIZATION (ROI)

KEY INSIGHT: MODERNIZING THE EXISTING CODEBASE VS. REBUILDING FROM ZERO

1. Retained Value in Current Codebase:
The existing repository already houses a fully engineered TypeScript backend, 11 responsive HTML5 page templates, a dual PostgreSQL/JSON persistence engine, an automated 21-test regression suite, and a working pdf-lib certificate generator. Rebuilding this from zero would needlessly expend Rs. 1,00,000 - Rs. 1,80,000 in redundant developer labor.

2. High-Impact Security Hardening Path:
Rather than paying for a complete rewrite, allocating Rs. 20,000 - Rs. 35,000 toward targeted hardening (locking down the public gallery upload endpoint, adding OTP/token-based password resets, and cryptographically validating Razorpay signatures) will instantly bring this system to institutional-grade security.

3. Production Hosting Advantage:
The application includes native Dockerfile and Nginx reverse proxy configurations, allowing it to be deployed on an affordable VPS (Rs. 600 - Rs. 1,200/month) rather than expensive proprietary SaaS platforms (which often cost Rs. 4,000 - Rs. 10,000/month for similar multi-role certificate management capabilities).

Report Prepared By: Lead Software Architect & Systems Security Engineer

For: Robotree Robotics LLP Executive Management & Strategic Planning

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